

AMAN AGARWAL

Delhi, India | amanagarwal0108@gmail.com | +91 9968823112 | LinkedIn

PROFESSIONAL SUMMARY

Senior AI Architect and Data Scientist with **9+ years** of experience building and shipping production AI systems at enterprise scale. Progressed from hands-on data science delivery to leading enterprise AI architecture — consistently taking on broader technical ownership, cross-functional leadership, and strategic AI decisions across global programmes at KPMG, Volvo, Floweraura, Decision Point, Publicis Sapient, and Fractal Analytics. Specialises in **Agentic AI, LLM engineering, RAG pipelines, and multi-agent orchestration** — with a strong record of translating complex business problems into governed, scalable AI products. Equally at home defining architecture strategy and writing production code. Has led cross-functional teams, authored **Architecture Decision Records**, shaped **AI technology roadmaps**, and championed **responsible AI** practices — from prompt injection defence to model fairness — across the full delivery lifecycle.

KEY SKILLS

Machine Learning

Python, PySpark, scikit-learn, Pandas, NumPy, Decision Trees, Regression, Classification, Ensemble Models, Random Forest, Boosting & Bagging, Clustering, Cascading, Time-Series Forecasting, Applied Mathematics, Predictive Maintenance, Association Rule Mining, Collaborative Filtering, A/B Testing, Reward Modeling, Human-in-the-Loop (HITL)

Deep Learning & NLP

PyTorch, TensorFlow, Keras, BERT, Transformers, Attention Models, LSTM, CNN, RNN, Encoder-Decoder Models, GAN, LLM, NLP, NLTK, SpaCy, Word Embeddings, Streaming Inference, Async Python

LLM & Generative AI

Generative AI, Prompt Engineering, Retrieval-Augmented Generation (RAG), Fine-Tuning LLMs, LangChain, LangGraph, Hugging Face Transformers, Azure OpenAI, OpenAI API, BM25, Hybrid Retrieval, Text Summarization, Conversational AI, Embedding Models, Vector Databases (FAISS, LanceDB, Milvus, Pgvector), Semantic Search, Context-Aware Retrieval, Tokenization & Attention, Prompt Tuning, Agentic Frameworks, Multi-Agent Orchestration, **Model Context Protocol (MCP)**, MCP Server Integration, Tool-Use Orchestration, Function Calling

LLM Evaluation & Observability

RAGAS (Retrieval Precision, Answer Faithfulness, Context Recall, Hallucination Rate), Langfuse (LLM Tracing, Prompt Versioning, Evaluation Tracking, Cost Monitoring), Domain-Specific Evaluation Frameworks, SQL Execution Accuracy, Schema Grounding Accuracy, Query Regeneration Rate, Token Efficiency, Agent Success Rate, Workflow Completion Rate, Feedback Loops, Output Versioning, Shadow Deployments

AI Architecture, Governance & Safety

Architecture Decision Records (ADRs), High-Level Design (HLD), AI Governance (Policy Frameworks, Audit Trails, Accountability Structures, Regulatory Compliance), EU AI Act Awareness, Responsible AI (Fairness, Transparency, Accountability), AI Guardrails (Input/Output Filtering, Content Moderation, Safety Layers, Policy Enforcement), AI Sandboxing (Isolated Agent Execution, Containerised Environments), OWASP LLM Top 10, Prompt Injection Defence, PII Redaction, AI Security & Access Control, Build vs Buy Evaluation, AI Technology Roadmap, Model Governance, Trusted AI Practices

Data Engineering & Analytics

SQL, RDBMS, Data Analytics, Data Processing, Pattern Recognition, Dimensionality Reduction (PCA, t-SNE), Anomaly Detection, Clustering Algorithms, Market Basket Analysis, Frequency Pattern Analysis, Elasticsearch, Apache Kafka, Snowflake, dbt, Geospatial Data, Power BI

Cloud, DevOps & MLOps

AWS (SageMaker, EC2, S3, Lambda, Kinesis), Azure (AKS, Databricks, Blob Storage, App Services, Bot Service, Application Insights), Microsoft Bot Framework, Microsoft Entra ID, Docker, Kubernetes, Terraform, FastAPI, Redis, ServiceNow REST APIs, CI/CD, MLOps (MLFlow), Prometheus, Grafana, SonarQube

PROFESSIONAL EXPERIENCE

Senior Data Scientist | *Fractal Analytics*

May 2026 – Present

Project: Agentic AI Platform for Enterprise IT Support (P&G)

- P&G's IT support was fragmented across manual channels with no unified AI layer; architected and shipped a production-grade **Agentic AI platform** integrating **Microsoft Teams**, **LangGraph**, and **OpenAI LLMs** with **ServiceNow** — eliminat-

ing manual IT support handling end-to-end and serving as the central AI backbone for enterprise support operations.

- Owned the full deployment stack using **Microsoft Bot Framework** and **Azure Bot Service** — implementing **Microsoft Entra ID** authentication, Teams app packaging, and structured release management across **development, staging, and production** environments, ensuring enterprise-grade security and compliance from day one.
- Support requests spanned knowledge queries, incidents, and catalog needs with no intelligent routing layer; designed a **Supervisor (Router) Agent** for **intent classification** and **multi-agent orchestration** — routing each request to specialised agents for **Knowledge Q&A**, **Incident Management**, and **Service Catalog**, fully automating three distinct workflows that previously required manual triage.
- Built three specialised AI agents — each with deep domain capability for its workflow — enabling end-to-end handling of support queries, automated **ServiceNow** incident creation via **MCP server** integration using **Model Context Protocol (MCP)**, and service catalog automation; complemented with **Redis-backed conversation memory** for seamless **multi-turn interactions** across sessions.
- To improve answer grounding and reduce hallucinations, built a **hybrid RAG pipeline** combining **BM25**, **FAISS vector search**, and **OpenAI embeddings**, underpinned by an offline **document indexing pipeline** for chunking, normalisation, and incremental vector updates — delivering citation-backed answers with low-latency **semantic search** across enterprise knowledge bases.
- To give agents native, structured access to enterprise systems, integrated **ServiceNow** as an **MCP server** using **Model Context Protocol (MCP)** — exposing incident creation, ticket updates, and catalog requests as **callable tools** that agents could invoke directly through **tool-use orchestration**, eliminating brittle raw REST calls; complemented with **Redis-backed conversation memory** for seamless **multi-turn interactions** across sessions.
- Established production **AI lifecycle pipelines** using **Langfuse** for **LLM tracing**, **prompt versioning**, and **evaluation tracking**; deployed agents in **sandboxed, isolated execution environments** on **Kubernetes (AKS)** — preventing unintended cross-agent data access and containing tool-use scope; wired **CI/CD** with automated rollback ensuring every release was reliable, reversible, and fully auditable.
- With no existing visibility into agent quality, defined **domain-specific evaluation frameworks** tracking **retrieval precision**, **answer faithfulness**, **hallucination rate**, **routing accuracy**, and **workflow completion rate** using **RAGAS**; built **observability dashboards** in **Azure Application Insights**, **Prometheus**, and **Grafana** — giving the team a live view of token usage, latency, and agent health that drove continuous prompt optimisation cycles.
- Owned the enterprise **AI governance** framework — establishing policy structures, audit trails, and accountability mechanisms across all AI pipelines; authored **ADRs** and **HLD documentation**, shaped the client's **AI technology roadmap**, and drove **build vs buy** decisions at leadership level; implemented **AI guardrails** for input/output filtering, content moderation, and safety policy enforcement alongside **OWASP LLM Top 10** mitigations, **prompt injection defence**, and **PII redaction**; mentored a cross-functional team of **5–10 engineers** on agentic design, **AI governance** practices, and **MCP-based tool integration** — accelerating the team's production AI capability.

Senior AI & LLM Engineer | *Publicis Sapient*

July 2024 – May 2026

Project: Enterprise NL2SQL Platform

- Enterprise analysts were waiting minutes for database query results with no natural language interface; led end-to-end architecture and delivery of a **Multi-Agent LLM system** using **LangChain** and **LangGraph** — from problem framing and schema design through evaluation and production rollout — cutting query turnaround to **10 seconds**, improving execution accuracy by **35%**, and serving as technical lead across product, governance, and engineering teams.
- Initial retrieval quality was causing frequent query regeneration; designed **hybrid Schema + Value Retrieval** with **Elasticsearch** and vector search, then instrumented **Langfuse** for **LLM tracing**, prompt versioning, and cost monitoring — driving systematic optimisation that cut regeneration cycles by **25%**.
- Latency was unacceptable and inference costs were escalating; built real-time **streaming inference** with **Async Python** and **FastAPI**, managed infrastructure as code via **Terraform**, and enforced **CI/CD** with **SonarQube** static analysis; deployed on **Kubernetes** with autoscaling — achieving **99.9%** uptime, **50%** lower latency, and **40%** reduction in inference cost.
- Without a feedback mechanism, prompt quality degraded silently in production; closed the learning loop via **feedback pipelines** capturing interaction logs and human corrections, tracking **SQL execution accuracy**, **schema grounding accuracy**, **query regeneration rate**, and **token efficiency** through Langfuse dashboards — enabling continuous, evidence-based prompt and retrieval improvement.
- Enforced enterprise security and multi-tenant governance; built **Prometheus** and **Grafana** dashboards tracking **query success rate**, **latency SLAs**, **token cost per query**, and **agent error rate** — giving the team a live, actionable view of production health that reduced reactive firefighting.

Project: LLM-Based Enterprise APIs (Attribute & Summary Extraction)

- Manual document processing was a bottleneck across enterprise workflows, with analysts spending hours extracting structured data from unstructured documents; designed and shipped production **FastAPI** services for structured **attribute extraction** and **document summarisation** (extractive & abstractive); implemented **schema-driven validation**, dynamic **prompt orchestration**, and async processing — cutting document processing time by over **70%** and reliably handling high-volume workflows from ingestion to structured JSON output at scale.

Project: Data Science Consulting for FMCG Clients

- FMCG clients were making critical pricing and promotional decisions without data-driven support; delivered end-to-end **ML solutions** for **Dynamic Pricing**, **Promotion Analytics**, and **Demand Planning** — turning commercial strategy into models that directly influenced pricing, trade, and supply chain decisions across major retail markets.
- Built **pricing optimisation models** using **price elasticity** and **cross-elasticity** analysis to improve margin and revenue; developed **promotion uplift** and **cannibalization models** that gave clients a clear, evidence-based view of trade promotion ROI — enabling smarter campaign planning and spend allocation.
- Supply chain planning was reactive and inventory allocation was inefficient; developed scalable **demand forecasting models** achieving **88% accuracy** — materially improving planning cycles and reducing inventory waste across multiple retail markets.
- Built distributed **data pipelines** on **PySpark**, **Azure Databricks**, and **AWS SageMaker** to handle large retail datasets efficiently across feature engineering, training, and batch inference — reducing pipeline run times and enabling faster model iteration.
- Owned the full **MLOps lifecycle** — CI/CD, deployment, monitoring, and model governance — and delivered **AI dashboards** that improved client business visibility by **20%**; led a team of **5–10 data scientists**, established engineering standards, and presented AI strategy directly to senior client stakeholders.

Data Scientist | Floweraura

December 2019 – August 2020

Project: Recommendation Engine Implementation

- Product discovery was purely generic with no personalisation for any user segment; designed and deployed a production **Recommendation Engine** using **Association Rule Mining** and **Collaborative Filtering**, serving **50K+ users** — delivering a **2%** revenue uplift within just 4 months of launch.
- Architected a **real-time serving layer** on **AWS** with **REST APIs** shipped via **Docker** and **GitHub Actions** CI/CD; optimised **feature engineering** pipelines for low-latency, high-throughput recommendation delivery — ensuring the engine scaled reliably under peak traffic.
- Ran **A/B tests** across user cohorts measuring **CTR and conversion lift** against a rule-based baseline; used **behavioral interaction data** to iteratively sharpen personalisation and targeting — validating model value and continuously improving recommendation quality; applied **data encryption** and AWS cloud security best practices throughout.

Data Scientist | Volvo Eicher

February 2019 – December 2019

Project: Telematics-Based Vehicle Maintenance System

- Vehicle component failures were unpredictable, causing unplanned downtime and costly reactive maintenance across fleet operations; took a **Telematics-based Predictive Maintenance** system from POC to production — building **real-time classification models** with strong **precision and recall** that cut unplanned downtime and maintenance costs, increasing deal conversion by **25%**.
- Telemetry data was arriving at high velocity with no structured ingestion layer; engineered **event-driven streaming pipelines** on **AWS Kinesis** for real-time telemetry processing; integrated **geospatial data** to optimise workshop dispatch and maintenance scheduling — turning raw vehicle signals into actionable field intelligence.
- Pushed predictive accuracy through advanced **feature engineering** on high-frequency sensor data; deployed a scalable **monitoring framework** for operational visibility; worked closely with business teams to translate model outputs into process changes — with adoption confirmed across field operations.

Career Break | Personal

November 2017 – February 2019

Took a planned career break to attend to a family medical emergency. Remained current with developments in data science and machine learning during this period.

Data Scientist | KPMG India

February 2016 – November 2017

Project: Predictive Maintenance for Enterprise Client

- The client was incurring significant costs from reactive asset maintenance with no predictive capability; built enterprise **Predictive Maintenance** models applying statistical modelling and **behavioral data analysis** — delivering **\$25M+** in cost savings and establishing one of the highest-impact early applications of ML in the client's operations.
- Performed exploratory data analysis and performance diagnostics across large-scale industrial datasets; distilled complex findings into clear, actionable recommendations for executive stakeholders — building lasting organisational confidence in data-driven decision making and securing continued investment in AI-led initiatives.

EDUCATION

Bachelor of Technology – Electronics and Communication Engineering
Dr. A.P.J. Abdul Kalam Technical University, India

July 2011 – August 2015

CERTIFICATIONS & ACHIEVEMENTS

- Data Science & Machine Learning Certification — Indian Institute of Technology (IIT), Delhi
- Data Science using Python — AnalytixLabs, India
- Awarded “**Rockstar Data Scientist**” at Decision Point Analytics for outstanding project delivery.
- Recognised with “**Kudos Award**” at KPMG India for driving measurable business impact.